

A quantitative refinement of a depth-three circuit lower bound

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Abstract

OpenAI’s depth-three circuit construction gives a single polynomial-time language whose n -bit membership function requires $2^{\omega(\sqrt{n})}$ total gates in unrestricted OR–AND–OR circuits. We derive the explicit lower bound $2^{c\sqrt{n \ln n / \ln \ln n}}$ for an absolute $c > 0$, at every sufficiently large input length, for the same language. The additional input is the published quantitative form of disjoint CNF sparsification, with clause density at most $(Kb)^{Kb}$ at width b . This permits the target width to grow as $\Theta(\ln n / \ln \ln n)$. We verify the counting, moment, restriction, and gate-count estimates uniformly for this schedule. We also reproduce the source’s finite restriction argument, including the backwards path count that removes dependence on the number of clauses. The refinement changes the proof parameters, not the language or its evaluation algorithm.

1 Introduction

For a Boolean function f , let $S_3(f)$ denote the minimum total number of AND and OR gates in an exact OR–AND–OR circuit for f . Bottom gates read input literals and optional constants; middle gates read bottom-gate outputs; and the output is an OR of middle-gate outputs. Fan-in and fan-out are unrestricted. Input negations are free, and all bottom, middle, and output gates are counted. Unary gates permit functions of smaller depth in this model.

OpenAI [1] constructs one language $L \in \mathsf{P}$ such that its n -bit membership function f_n satisfies $\log_2 S_3(f_n)/\sqrt{n} \rightarrow \infty$. Its argument first fixes a desired constant in the circuit exponent, then chooses a constant CNF width, and finally takes the input length sufficiently large. The present note extracts an explicit rate by allowing that width to grow. The language is recalled completely in Section 3.

Theorem 1.1. *For the language L of Definition 3.1, there exist absolute constants $c > 0$ and n_0 such that*

$$S_3(f_n) > 2^{c\sqrt{n \ln n / \ln \ln n}} \quad \text{for every } n \geq n_0. \quad (1.1)$$

The language is decided by one deterministic polynomial-time algorithm. For any absolute $K \geq 2$ satisfying Lemma 2.1, one may take $c = 1/(6144\sqrt{K})$.

The displayed value of c is explicit in terms of K ; no numerical value of the sparsification constant is claimed. The threshold n_0 is absolute once this constant has been fixed.

Contribution and attribution. The language, clause-count-independent restriction lemma, and circuit contradiction are from [1]. The restriction proof reproduced in Appendix A is an adaptation of that source, not a new restriction lemma. The extra ingredient is the quantitative

sparsification estimate of Calabro, Impagliazzo, and Paturi [2], combined with the disjoint refinement of Dell et al. [3]; Williams [4] explicitly uses this quantitative counting form. Our contribution is the growing-parameter schedule and the uniform estimates that yield (1.1) for the existing construction.

The distinction between unrestricted and bounded-bottom-fan-in circuits is essential. Recent exponential bounds for majority and inner product concern restricted models [6, 7]. The stronger unrestricted majority bound $2^{\Omega(\sqrt{n \log n})}$ is presented as a conjectural target in [6]. The rate in (1.1) is not stated in the source manuscript or its accompanying formalization. A targeted literature search found no prior statement of this quantitative refinement; this observation does not establish publication priority.

Proof overview. The construction has data bits and parameter bits. Suitable fixed parameter bits give a data slice whose sign has small correlation with every narrow CNF on most random restrictions. Sparsification makes the family of tests small enough for a limited-independence moment bound. The restriction lemma transfers this correlation bound to wider CNFs. After deleting very wide bottom clauses from a hypothetical small circuit, each middle gate accepts too few inputs to cover the slice’s positive set. Quantitative sparsification permits width $b = \Theta(\ln n / \ln \ln n)$ and hence a circuit exponent of order $\sqrt{bn}$.

2 Correlation and the two structural tools

The cube $\{0, 1\}^V$ on a finite coordinate set V has uniform measure. Boolean functions are identified with 0/1 indicators. A clause is an OR of literals, and its width is the number of distinct variables after normalization. A false clause has width zero; an empty CNF is true. For $b \geq 1$ and a real function G on this cube, define

$$\text{corr}_b(G) = \max_J |\mathbb{E}[GJ]|,$$

where J ranges over all width-at-most- b CNF indicators, without a clause-count restriction. Constants are allowed. There are finitely many such indicators, so the maximum exists.

A p -random restriction retains each coordinate independently with probability p , giving a live set T , and assigns the other coordinates uniformly, giving ρ . Write $G_{T,\rho}$ for the remaining function. Uniform completion of a random restriction is uniform on the full cube; therefore

$$\mathbb{E}_x G(x) H(x) = \mathbb{E}_{T,\rho} \mathbb{E}_z G_{T,\rho}(z) H_{T,\rho}(z). \quad (2.1)$$

Lemma 2.1 (Quantitative disjoint sparsification). *There exist an absolute constant $K \geq 2$ and positive integers $M(b)$, for $b \geq 2$, such that every width- b CNF indicator on $m \geq 1$ variables is a sum of at most $2^{m/8}$ pairwise disjoint width- b CNF indicators, each having at most $M(b)m$ clauses, with*

$$M(b) \leq (Kb)^{Kb}. \quad (2.2)$$

Derivation from quantitative sparsification. We use the quantitative counting form stated in [4, Section 4, p. 6:9], combining [2, Section 6] with the disjoint refinement in [3, Appendix A, Lemma A.1 and its following proof]. Fix an absolute error $\eta_0 \in (0, 1/8]$ in that bound. The quantitative estimate $(b/\eta_0)^{O(b)}$ means that there are absolute constants $A, C \geq 1$, independent of both b and m , for which the density before adding branch-defining units is at most

$$D(b) \leq A(b/\eta_0)^{Cb}.$$

The finite sparsification recursion has at most $2^{\eta_0 m}$ leaves for every input on m variables. Lemma A.1 of [3] states these finite quantifiers explicitly; they are bounds on the leaves and formulas, not just an asymptotic running-time estimate.

Here is the disjointness modification. At a flower with heart clause $H = \bigvee_{\ell \in U} \ell$ and petals P_i , the relevant clauses are $H \vee P_i$. The heart branch enforces H ; the petal branch enforces all P_i and the units $\neg \ell$ for $\ell \in U$. These two branches partition the satisfying assignments according to the truth of H . Iterating gives disjoint leaves. A consistent leaf contains at most one distinct added unit per variable, hence at most m ; inconsistent leaves are discarded. Width cannot increase. The reduction step removes clauses properly containing a unit, so these units create no new flowers and the leaf bound is unchanged, as proved in [3, Appendix A].

Set $M(b) = \lceil D(b) \rceil + 1$. For example, choosing any absolute $K \geq \max\{2, 2C, \eta_0^{-1}, 3A\}$ gives

$$M(b) \leq D(b) + 2 \leq 3A(b/\eta_0)^{Cb} \leq (Kb)^{Kb} \quad (b \geq 2).$$

Also $2^{\eta_0 m} \leq 2^{m/8}$. Both estimates hold for every finite m , including when $b > m$. The identically false formula may be represented by no nonzero leaves. This proves the claimed indicator identity. No numerical values of A, C, η_0 , or K are needed or asserted here. $\square$

Lemma 2.2 (Restriction bound from [1]). *Let $k \geq 0$ and $b \geq 1$ be integers, let H be a width- k CNF on d variables, and let $0 < p < 1$. If*

$$\theta = \sum_{\ell=b+1}^k \binom{k}{\ell} \left(\frac{2p}{1-p} \right)^\ell < 1, \quad (2.3)$$

there is a nonnegative $W_H = W(H, T, \rho)$ such that

$$\mathbb{E} W_H \leq (1 - \theta)^{-1}, \quad |\mathbb{E}_z G(z) H_{T, \rho}(z)| \leq W_H \text{corr}_b(G)$$

simultaneously for every real function G on the live cube.

The full finite proof appears in Appendix A. It applies to each finite choice of b, k, d, p ; no fixed-width limiting argument is part of the lemma.

3 The language and finite hard-slice estimates

We recall the evaluator from [1, Section 4]. Its use of parameter bits to select hard subfunctions is related to the input-index method of Paturi, Saks, and Zane [5, Section 5].

Definition 3.1 (The fixed language). For input length n , set $d = \lfloor n/5 \rfloor$ and reject if $d < 64$. Otherwise set

$$r = \lceil d^{2/3} \rceil, \quad t = \max\{j \geq 0 : j \text{ is even and } j^6 \leq d\}.$$

As a safety check, reject if fewer than $2d + (t + 2)r - 1$ bits are available; Proposition 3.2 shows this branch is never taken for $d \geq 64$. Parse the input, in order, into d data bits x , $d + r - 1$ hash bits u , r modulus bits $p_0, \dots, p_{r-1}$, and t coefficient blocks of r bits each. Ignore unused bits. The modulus bits define

$$P(Z) = Z^r + \sum_{i=0}^{r-1} p_i Z^i, \quad R_P = \mathbb{F}_2[Z]/(P),$$

and the coefficient blocks define $\beta_0, \dots, \beta_{t-1} \in R_P$ in the basis $1, Z, \dots, Z^{r-1}$. Let

$$h_i(x) = \sum_{a=1}^d u_{i+a-1} x_a \pmod{2} \quad (1 \leq i \leq r), \quad h_u(x) = \sum_{i=1}^r h_i(x) Z^{i-1}. \quad (3.1)$$

If $\lambda : R_P \rightarrow \mathbb{F}_2$ selects the constant coefficient of the representative of degree less than r , accept exactly when

$$\lambda \left(\sum_{j=0}^{t-1} \beta_j h_u(x)^j \right) = 0. \quad (3.2)$$

Proposition 3.2. *Definition 3.1 gives one deterministic polynomial-time language. For all sufficiently large d ,*

$$r \geq d^{2/3}, \quad \frac{1}{2} d^{1/6} \leq t \leq d^{1/6}. \quad (3.3)$$

Proof. For $d \geq 64$, $r \leq 2d^{2/3}$ and $t+2 \leq 2d^{1/6}$. Thus the blocks use $2d + (t+2)r - 1 \leq 2d + 4d^{5/6} - 1 \leq 4d - 1 \leq n$ bits. Also $d^{1/6} - 2 < t \leq d^{1/6}$, which gives (3.3) for $d \geq 4096$. Already $t \geq 2$ for $d \geq 64$, which is all that the finite criterion below requires. The integer parameters can be found by testing $r^3 \geq d^2$ and the sixth powers of even integers, with polynomial overhead.

Every monic modulus gives a well-defined quotient ring, whether or not it is irreducible. School-book multiplication and monic reduction cost $O(r^2)$ bit-array operations; the hash costs $O(rd)$; and Horner evaluation costs $O(tr^2)$. These fixed loops are implementable by one Turing machine with polynomial overhead. The evaluator neither searches for hard parameters nor tests irreducibility. $\square$

Lemma 3.3 (Finite hard-slice criterion). *Let $d \geq 64$, with r, t as above, and let $b \geq 2$, $0 < p < 1$, and $\mu = pd \geq 2$. Set $M = M(b)$ from Lemma 2.1. Assume $r \geq (33/16)\mu$ and, for every integer $m \in [\mu/2, 2\mu]$,*

$$\frac{M[1 + b \log_2(2m+1)]}{t} + \frac{\log_2 t}{m} \leq \frac{1}{8}. \quad (3.4)$$

There is a fixed setting of the non-data bits whose sign $g : \{0, 1\}^d \rightarrow \{-1, 1\}$, with $+1$ on acceptance, satisfies

$$\mathbb{P}_{T,\rho} \left[|T| \notin [\mu/2, 2\mu] \text{ or } \text{corr}_b(g_{T,\rho}) > 2^{-|T|/8} \right] \leq 4 \cdot 2^{-\mu/16}. \quad (3.5)$$

Proof. Choose a monic irreducible P of degree r , which exists over $\mathbb{F}_2$ for each $r \geq 1$, and identify R_P with $F = \mathbb{F}_{2^r}$. Choose u and the coefficients β_j independently and uniformly. We first bound the joint failure probability over these choices and the random restriction.

Hash injectivity. For nonzero $w \in \{0, 1\}^d$, let a be its largest nonzero coordinate. The i th output bit of $h_u(w)$ contains u_{i+a-1} with coefficient one, and all other indices in that row are smaller. These successive pivots show that the linear map $u \mapsto h_u(w)$ has rank r . Hence its image is uniform and $\mathbb{P}_u[h_u(w) = 0] = 2^{-r}$. A restricted data cube of dimension m has $2^m - 1$ nonzero difference vectors, all supported on T . Linearity and a union bound give

$$\mathbb{P}_u[h_u \text{ is not injective on this cube}] \leq (2^m - 1)2^{-r} \leq 2^{m-r}. \quad (3.6)$$

Limited independence. Fix an injective hash on the cube. For any $q \leq t$ distinct points, the hashes are distinct elements $a_1, \dots, a_q \in F$. Lagrange interpolation shows that evaluation of degree-less-than- t polynomials onto F^q is surjective. Uniform coefficients therefore give independent uniform values at those points. Since λ is a nonzero binary linear functional, applying $(-1)^\lambda$ yields unbiased t -wise independent signs. Thus the following moment bound is conditional on any injective hash.

Sparse tests. There are at most $(2m+1)^b$ normalized clauses of width at most b : encode their literals in a fixed order and pad to length b . Include the empty false clause. Lists of Mm clauses, with an extra padding symbol, then encode at most

$$2^{Mm[1+b\log_2(2m+1)]} \quad (3.7)$$

CNF indicators. This includes both constants.

For one such indicator J , put $Y = \sum_{z \in \{0,1\}^T} g_{T,\rho}(z)J(z)$. In the expansion of Y^t , a tuple with a point occurring exactly once has zero expectation. Every remaining tuple uses at most $t/2$ distinct points. Listing $t/2$ points, with padding, and a choice of list position for each factor gives at most $t^t 2^{mt/2}$ tuples. Since t is even,

$$\mathbb{E}|Y|^t \leq t^t 2^{mt/2}, \quad \mathbb{P}[|\mathbb{E}_z g_{T,\rho} J| > 2^{-m/4}] \leq t^t 2^{-mt/4}.$$

Combining this with (3.7), the probability that any sparse test fails is at most

$$2^{Mm[1+b\log_2(2m+1)]+t\log_2 t - mt/4} \leq 2^{mt(1/8-1/4)} = 2^{-mt/8},$$

where the inequality is exactly (3.4). On the complementary event, Lemma 2.1 expresses every width- b CNF as a sum of at most $2^{m/8}$ tests from this same sparse family. The triangle inequality gives $\text{corr}_b(g_{T,\rho}) \leq 2^{-m/8}$ simultaneously for all width- b CNFs.

Restriction tails and averaging. The live dimension m is binomial with mean μ . Independence gives

$$\mathbb{E}2^{-m} = (1-p/2)^d \leq e^{-\mu/2}, \quad \mathbb{E}2^m = (1+p)^d \leq e^\mu.$$

Markov's inequality yields

$$\begin{aligned} \mathbb{P}[m < \mu/2] &\leq 2^{-((\log_2 e - 1)/2)\mu}, \\ \mathbb{P}[m > 2\mu] &\leq 2^{-(2-\log_2 e)\mu}. \end{aligned}$$

Both exponent coefficients exceed $1/16$, so their sum is at most $2 \cdot 2^{-\mu/16}$. Inside the dimension window, $r \geq (33/16)\mu$ gives $2^{m-r} \leq 2^{-\mu/16}$, while $t \geq 2$ gives $2^{-mt/8} \leq 2^{-\mu/16}$. Adding the two errors and the two dimension tails bounds joint failure by $4 \cdot 2^{-\mu/16}$. Averaging over the finitely many settings of $(u, \beta_0, \dots, \beta_{t-1})$ leaves one setting with at most this failure probability over restrictions. Fix it, the chosen P , and arbitrary unused bits. $\square$

4 Growing parameters and the lower bound

We now prove Theorem 1.1. All unqualified logarithms are natural. For sufficiently large d , set

$$\begin{aligned} b &= \left\lfloor \frac{\ln d}{24K \ln \ln d} \right\rfloor, & s &= \frac{\sqrt{b}}{256}, & B &= 64(s+1), \\ p &= \frac{B}{\sqrt{d}}, & \mu &= B\sqrt{d}, & k &= \left\lceil 3(s+1)\sqrt{d} \right\rceil. \end{aligned}$$

We work beyond a fixed absolute threshold so that $b \geq 65536$, $s \geq 1$, and $p \leq 1/2$. Let $M = M(b)$ from Lemma 2.1.

The sparse-test union bound. Eventually $\ln(Kb) \leq 2 \ln \ln d$, so (2.2) gives

$$\ln M \leq Kb \ln(Kb) \leq \frac{\ln d}{12}, \quad M \leq d^{1/12}.$$

Uniformly for $\mu/2 \leq m \leq 2\mu$, we have $\log_2(2m+1) = O(\ln d)$ and $b = O(\ln d)$. More explicitly, (3.3) gives the finite envelope

$$\begin{aligned} \frac{M[1 + b \log_2(2m+1)]}{t} + \frac{\log_2 t}{m} &\leq 2Md^{-1/6}[1 + b \log_2((4B+1)\sqrt{d})] + \frac{\ln d}{3B \ln 2 \sqrt{d}} \\ &\leq 2d^{-1/12}[1 + b \log_2((4B+1)\sqrt{d})] + \frac{\ln d}{3B \ln 2 \sqrt{d}}. \end{aligned}$$

Consequently,

$$\frac{M[1 + b \log_2(2m+1)]}{t} + \frac{\log_2 t}{m} = O\left(d^{-1/12}(\ln d)^2 + \frac{\ln d}{\sqrt{d}}\right) = o(1).$$

The constants here are absolute once the absolute K is fixed. This proves (3.4) uniformly over the entire window.

Hash and restriction tails. Since $B = O(\sqrt{\ln d})$, we have $r/\mu \rightarrow \infty$. In particular, eventually $r \geq (33/16)\mu$. Within the window,

$$2^{m-r} \leq 2^{-\mu/16}, \quad 2^{-mt/8} \leq 2^{-\mu/16}.$$

For the binomial random variable $m = |T|$ of mean μ , the exponential-moment bounds in Lemma 3.3 give

$$\mathbb{P}[m \notin [\mu/2, 2\mu]] \leq 2 \cdot 2^{-\mu/16}.$$

Averaging the joint failure probability over the parameter bits therefore gives one setting of those bits, for this length, whose sign function g satisfies

$$\mathbb{P}[m \notin [\mu/2, 2\mu] \text{ or } \text{corr}_b(g_{T,\rho}) > 2^{-m/8}] \leq 4 \cdot 2^{-\mu/16}. \quad (4.1)$$

This is a pointwise existence argument at each d ; it does not exchange the original fixed-parameter asymptotic quantifiers.

The restriction lemma still applies. For $d \geq 1$,

$$k \leq [3(s+1) + 1]\sqrt{d}, \quad \frac{2pk}{1-p} \leq 4B[3(s+1) + 1] =: Q.$$

Using $B = 64(s+1)$ and $s \geq 1$,

$$Q \leq 1024(s+1)^2 \leq 4096s^2 = \frac{b}{16}.$$

By $\binom{k}{\ell} \leq k^\ell/\ell!$ and $\ell! \geq (\ell/e)^\ell$,

$$\theta \leq \sum_{\ell > b} \frac{Q^\ell}{\ell!} \leq \sum_{\ell > b} (e/16)^\ell < \frac{1}{2}.$$

Thus Lemma 2.2 supplies $\mathbb{E}W_H \leq 2$ for every width- k test H .

4.1 Correlation transfer and the circuit contradiction

On the good restrictions in (4.1), the restricted correlation is at most $W_H 2^{-\mu/16}$. On the bad restrictions it is at most one. Uniform completion of a random restriction is uniform on $\{0,1\}^d$. Taking expectations and using the triangle inequality gives

$$|\mathbb{E}_x g(x)H(x)| \leq 6 \cdot 2^{-\mu/16} \quad (4.2)$$

for every width- k CNF H . Independence of W_H and the good event is not required. Applying (4.2) to the constant-true CNF shows that $\mathbb{P}[g = 1] \geq 1/4$ for sufficiently large d .

Suppose there were a circuit for f_n with at most $S \leq 2^{s\sqrt{d}}$ total gates. Fix its parameter inputs to the setting chosen above. Delete gates not connected to the output. Each remaining middle AND gate is a CNF H with at most S distinct bottom clauses, and there are at most S middle gates, even with shared bottom gates. Normalize its clauses by deleting true clauses and repeated literals; tautological clauses are true and are removed. A middle gate with a false clause computes zero and may be omitted. Delete from H the clauses of width greater than k , obtaining H' . Then $H' \geq H$ and

$$\mathbb{E}(H' - H) \leq S2^{-k}.$$

Every input accepted by H is accepted by the top OR, so $gH = H$. Equation (4.2) therefore implies

$$\mathbb{E}H \leq 6 \cdot 2^{-\mu/16} + S2^{-k}.$$

Taking the union over middle gates yields

$$\begin{aligned} \mathbb{P}[g = 1] &\leq 6S2^{-\mu/16} + S^2 2^{-k} \\ &\leq 6 \cdot 2^{-(3s+4)\sqrt{d}} + 2^{-(s+3)\sqrt{d}} = o(1), \end{aligned}$$

contradicting $\mathbb{P}[g = 1] \geq 1/4$. Hence $S_3(f_n) > 2^{s\sqrt{d}}$ for every sufficiently large n . Finally,

$$s\sqrt{d} = \frac{\sqrt{bd}}{256} = \Theta\left(\sqrt{\frac{n \ln n}{\ln \ln n}}\right),$$

because $d = \lfloor n/5 \rfloor$ and $b = \Theta(\ln d / \ln \ln d)$. More explicitly, eventually

$$d \geq \frac{n}{6}, \quad \ln d \geq \frac{\ln n}{2}, \quad b \geq \frac{\ln d}{48K \ln \ln d} \geq \frac{\ln n}{96K \ln \ln n}.$$

It follows that

$$s\sqrt{d} \geq \frac{1}{6144\sqrt{K}} \sqrt{\frac{n \ln n}{\ln \ln n}},$$

which proves Theorem 1.1. □

5 Remarks on scope

The gain in the exponent is the unbounded factor $\sqrt{\ln n / \ln \ln n}$. It remains smaller than every fixed positive power of n , so this argument does not give $2^{n^{1/2+\varepsilon}}$ for fixed $\varepsilon > 0$. The role of quantitative sparsification is specific: $M(b) \leq \exp(O(b \log b))$ remains small enough compared with the $d^{1/6}$ independence parameter when b is of order $\ln d / \ln \ln d$. The relation $s = \sqrt{b}/256$ then converts this width into the asserted circuit exponent.

Size of the threshold. The proof is asymptotic, with extremely conservative constants. The requirement $b \geq 65536$ alone implies

$$\frac{\ln d}{\ln \ln d} \geq 24K \cdot 65536 \geq 3,145,728.$$

On the large- d branch this requires d to have more than 24 million decimal digits. The other eventual inequalities may increase the threshold further. This is a limitation of the displayed parameter schedule, not a claim about the smallest lengths at which the lower bound is true.

The logarithmic loss. With the available estimate $M(b) \leq \exp(O(b \ln b))$, certifying that $M(b)$ stays within a fixed power of d leads to $b \ln b = O(\ln d)$. This accounts for the $\ln \ln n$ loss. A hypothetical uniform disjoint sparsification theorem with $M(b) \leq 2^{Cb}$ for an absolute C and the same $2^{m/8}$ piece bound would permit $b = \delta \ln d$ for a sufficiently small absolute $\delta > 0$. The same estimates would then give $2^{\Omega(\sqrt{n \ln n})}$ for this language, without implying the conjectural majority bound. Section 7 of [2] asks about a corresponding improvement in sparsification density. We do not assert that such a disjoint improvement is impossible; the present density estimate supplies a sufficient bound, not a matching lower bound.

Increasing t to another fixed power of d , while maintaining the input budget and hash estimates, would enlarge the polynomial density allowance. With the same sparsification estimate it would change constants, not the order $\ln d / \ln \ln d$ of the certified width. Moreover, t determines the number of coefficient blocks, so changing it would define a variant of the language rather than merely retune its proof.

All choices of parameter bits are existential restrictions of one fixed language. They may depend on n and on the proof parameters, while the membership algorithm does not. The proof uses no uniformity assumption on the competing circuits.

A The finite restriction argument

For completeness, we reproduce the argument of [1, Section 2] in the notation of Lemma 2.2. The two points to check are an exact finite signed expansion into width- b tests and a bound on its expected absolute coefficient mass. Counting paths backwards is what removes the number of clauses from that bound.

Proof of Lemma 2.2. Normalize H by deleting tautological and true clauses and deleting repeated literals within each remaining clause. Keep clause indices, including repeated clauses. Write E_i for the scope of clause C_i . A violation of C_i specifies a unique pattern on E_i ; the false clause has empty scope. Fix a restriction (T, ρ) , and let $v_i(z)$ be its violation indicator. For $U \subseteq T$ set

$$q = \sum_i v_i, \quad a_U = \sum_{i: |E_i \cap (T \setminus U)| \leq b} v_i, \quad c_U = \sum_{i: E_i \cap (T \setminus U) = \emptyset} v_i.$$

Call an index hard at U if $|E_i \cap (T \setminus U)| > b$. The function $K_0 = \mathbf{1}[a_\emptyset = 0]$ is a width- b CNF.

A path $P = (i_1, \dots, i_j)$ starts at $U_0 = \emptyset$, selects an index hard at U_{h-1} , and sets $U_h = U_{h-1} \cup (E_{i_h} \cap T)$. Each step adds at least $b+1$ coordinates, so paths have length at most $L = \lfloor |T|/(b+1) \rfloor$. On the support of $\prod_{h=1}^j v_{i_h}$, the selected indices are distinct and $a_{U_h} \geq c_{U_h} \geq h$. All fractions below are defined as zero when their numerator vanishes.

For a nonempty path define

$$r_P = \frac{\prod_{h=1}^j v_{i_h}}{q \prod_{h=1}^{j-1} a_{U_h}}, \quad t_P = \frac{\prod_{h=1}^j v_{i_h}}{\prod_{h=1}^j a_{U_h}}.$$

Since $q = a_{U_j} + \sum_{i \text{ hard at } U_j} v_i$,

$$r_P = t_P - \sum_{i \text{ hard at } U_j} r_{Pi}.$$

At a point with $K_0 = 1$ and $q > 0$, all violated clauses are initially hard, and $\sum_{|P|=1} r_P = 1$. Iterating the displayed identity until there are no further paths gives

$$H_{T,\rho} = K_0 + \sum_{j=1}^L (-1)^j \sum_{|P|=j} K_0 t_P. \quad (\text{A.1})$$

At points with $K_0 = 0$ every term vanishes; at points with $q = 0$ the nonempty-path terms vanish and $K_0 = H_{T,\rho} = 1$. Thus this is an identity on the whole cube, not just on its error set.

Call a path compatible with ρ when its selected violating patterns are mutually consistent and agree with ρ on fixed coordinates. For such a path, all selected violations prescribe a single pattern σ_P on U_j . Denote its cylinder by Z_P . The numerator of t_P is $\mathbf{1}_{Z_P}$. On Z_P the function c_{U_h} is a positive constant c_h , because all variables of every counted clause have been fixed. Moreover,

$$a_{U_h} = c_h + w_h, \quad w_h = \sum_{i: 0 < |E_i \cap (T \setminus U_h)| \leq b} v_i.$$

After substituting the final pattern σ_P , the indexed clauses counted by w_h have width at most b on the remaining coordinates. Their multiplicities and any constant clauses are retained.

For any indexed family of clauses and positive integer c , randomly permute its clause objects together with c distinct markers, and take the conjunction of the clauses preceding the first marker. On an assignment violating w of those clauses, this conjunction holds with probability $c/(c+w)$: the earliest of the w violated clause objects and c markers must be a marker. This is a finite convex mixture of CNF indicators. Apply the construction independently for each h , then conjoin the sampled CNFs with K_0 and the unit clauses defining Z_P . Every resulting test J_P has width at most b , using $b \geq 1$. It satisfies

$$K_0 t_P = \omega(P) \mathbb{E} J_P, \quad \omega(P) = \prod_{h=1}^j \frac{1}{c_h}.$$

An incompatible path contributes zero. Define

$$W_0 = 1, \quad W_j = \sum_{\substack{|P|=j \\ P \text{ compatible with } \rho}} \omega(P), \quad W = \sum_{j=0}^{\lfloor d/(b+1) \rfloor} W_j.$$

Equation (A.1) and the triangle inequality prove $|\mathbb{E}_z G H_{T,\rho}| \leq W \text{corr}_b(G)$ for every G simultaneously. The construction of W never uses G .

It remains to prove $\mathbb{E} W_j \leq \theta^j$ for $j \geq 1$. Generate a uniform full assignment y , independently of T , and set $\rho = y|_{[d] \setminus T}$. Conditional on (T, ρ) , a compatible path has all its clauses violated by y with probability $2^{-|U_j|}$. Consequently

$$\mathbb{E} W_j = \mathbb{E}_{T,y} \sum_{\substack{|P|=j \\ C_{i_1}, \dots, C_{i_j} \text{ violated by } y}} 2^{|U_j|} \prod_{h=1}^j \frac{1}{c_{U_h}(y|_T)}. \quad (\text{A.2})$$

This is conditional cylinder averaging; it makes no claim of independence between the path and its denominators.

Fix y , and for a remainder $R \subseteq [d]$ put

$$V_R = \{i : C_i(y) = 0, E_i \cap R = \emptyset\}.$$

An actual path has $R_h = T \setminus U_h$ and $D_h = U_h \setminus U_{h-1}$, with

$$i_h \in V_{R_h}, \quad D_h \subseteq E_{i_h}, \quad |D_h| > b, \quad R_{h-1} = R_h \cup D_h, \quad c_{U_h}(y|T) = |V_{R_h}|.$$

Its terminal remainder R_j and the reverse sequence $(i_j, D_j), \dots, (i_1, D_1)$ determine both T and its entire indexed path. Hence this is an injective encoding even when clauses repeat.

Let $\alpha = 2p/(1-p)$ and $\pi(R) = p^{|R|}(1-p)^{d-|R|}$. The weight of an encoded path in (A.2), including its live-set probability, is exactly

$$\pi(T)2^{|U_j|} \prod_h \frac{1}{|V_{R_h}|} = \pi(R_j) \prod_h \frac{\alpha^{|D_h|}}{|V_{R_h}|}.$$

Enlarge the reverse sequences by allowing, at any state R , every choice $i \in V_R$ and $D \subseteq E_i$ with $|D| > b$, followed by $R \cup D$. Here $D \cap R = \emptyset$ because $i \in V_R$. All weights are nonnegative, and the outgoing weight at each state with $V_R \neq \emptyset$ is at most

$$\frac{1}{|V_R|} \sum_{i \in V_R} \sum_{\ell=b+1}^{|E_i|} \binom{|E_i|}{\ell} \alpha^\ell \leq \sum_{\ell=b+1}^k \binom{k}{\ell} \alpha^\ell = \theta.$$

If V_R is empty there are no outgoing steps. Induction on the number of reverse steps bounds their total weight from any state by θ^j . Summing over terminal remainders costs only $\sum_R \pi(R) = 1$. This proves $\mathbb{E}_T[\text{path mass}] \leq \theta^j$ for every fixed y , and therefore $\mathbb{E}W_j \leq \theta^j$.

Finally $\mathbb{E}W \leq \sum_{j=0}^{\lfloor d/(b+1) \rfloor} \theta^j \leq (1-\theta)^{-1}$. Empty formulas, false clauses, repeated clauses, empty live sets, and $k \leq b$ all satisfy the same argument. $\square$

B Verification and provenance

The source construction and restriction argument were read at repository revision `adc7f1241b42e`. The accompanying research note records the detailed literature comparison and search scope. The manuscript date of the source is September 23, 2026; its inspected Git revision is dated October 6, 2026. These dates alone do not establish publication priority for an unstated consequence.

Two Python programs accompany the derivation. The file `check_quantitative_depth_three.py` checks the elementary constants with exact arithmetic and nine illustrative parameter choices in the log domain. The chosen sample values of K are not certified sparsification constants. The file `check_restriction_lemma.py` checks all 512 sets of normalized clauses on two variables, and 11 additional formulas at two rational retention probabilities. Its 534 formula-parameter cases cover 11,254 restrictions and 43,524 completions, including nonzero three-step paths and marker mixtures with seven indexed clauses. All checks passed. These finite diagnostics support the proof but do not replace its universal arguments.

The repository contains a Lean theorem for the restriction lemma and finite numerical estimates. The strengthened rate and growing schedule in this paper have not been integrated into that formalization, and no fresh Lean kernel verification was performed for this work. The manuscript was prepared with assistance from Codex, including derivation review, literature search, computational diagnostics, and typesetting. It is a research preprint and has not undergone external peer review.

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